Transformer Design for Isolated DC-DC Converter-II

Introduction

In the rapidly evolving landscape of charging infrastructure, the efficiency and reliability of power converters are paramount. At the heart of many DC-DC converters, especially isolated topologies, lies the transformer—a critical component responsible for voltage transformation and galvanic isolation. However, designing an optimal transformer is a complex task, balancing various electrical and physical parameters to minimize losses and ensure compact, cost-effective operation. This lecture delves into the intricate details of transformer design, building upon foundational principles to explore how physical dimensions, material properties, and winding configurations collectively impact performance, ultimately guiding engineers toward an optimized design for high-frequency applications. Understanding these principles is crucial for developing robust and efficient power solutions for modern charging systems.

Core Concepts

Term	Definition	Significance
`Core Loss`	Energy dissipated within the transformer's magnetic core, primarily due to hysteresis and eddy currents, which manifest as heat.	Directly impacts transformer efficiency and thermal management; depends on core material, flux density, and frequency.
`Winding Loss (Copper Loss)`	Energy dissipated in the transformer windings due to the resistance of the conductor material, proportional to the square of the current.	A major contributor to overall transformer losses, affecting efficiency and requiring careful wire sizing and winding arrangement.
`Window Area (WA)`	The physical space within the transformer core structure where the windings are placed.	Determines the maximum volume available for conductors and insulation,

Term	Definition	Significance
		influencing the number of turns and wire gauge.
`Mean Length Per Turn (MLT)`	The average length of a single turn of wire around the transformer core.	Essential for calculating the total length of wire in a winding, which directly affects its DC resistance and copper loss.
`Winding Fill Factor (Ku)`	A dimensionless factor representing the ratio of the actual copper area to the total available window area.	Accounts for the non-ideal packing of circular wires and insulation, indicating how effectively the window area is utilized.
`Peak Flux Density (ΔB)`	The maximum change in magnetic flux density within the transformer core during operation.	A critical parameter influencing both core losses and the number of turns required; must be kept below saturation limits.
`Fraction of Window Area (γ_j)`	The proportion of the total window area allocated to a specific winding `j`.	Allows for the distribution of available winding space among multiple windings, crucial for optimizing copper losses.
`Turns Ratio`	The ratio of the number of turns in one winding to the number of turns in another (e.g., secondary to primary).	Defines the voltage transformation ratio and is fundamental for achieving desired output voltages from the converter.
`Optimization`	The process of systematically adjusting design parameters (e.g., `γ_j`, `ΔB`) to achieve a desired performance goal, such as minimal total losses.	Ensures the transformer operates at peak efficiency, minimizing heat generation and maximizing power delivery.

Detailed Analysis

1. Recap of Transformer Losses and Core Characteristics

The design of a transformer begins with understanding the losses associated with its operation. These are broadly categorized into core losses and winding losses (also known as copper losses).
* **Core Losses**: These occur in the magnetic core material and consist of:
* **Hysteresis Losses**: Energy dissipated due to the repeated magnetization and demagnetization of the core material.
* **Eddy Current Losses**: Losses caused by circulating currents induced within the core material by changing magnetic fields.
* **Core Loss Measurement**: Since hysteresis and eddy current losses are difficult to measure separately, manufacturers typically provide core loss data in terms of "core loss per unit volume" (e.g., watts per cubic centimeter). This data is often presented as logarithmic curves in material datasheets, plotting core loss against peak AC flux density (ΔB) for various operating frequencies.
* The general expression for core loss is given by: `P_core = K_core * (ΔB)^β * A_c * l_c`, where `K_core` is a constant dependent on frequency, `ΔB` is the peak flux density, `β` is a material property (typically 2.5 to 2.7), `A_c` is the core cross-sectional area, and `l_c` is the magnetic path length (where `A_c * l_c` represents the core volume).
* **Winding Losses (Copper Losses)**: These occur in the conductive windings due to their inherent resistance. They are calculated as `I^2 * R`, where `I` is the RMS current and `R` is the winding resistance.

2. Key Design Outcomes and Peak Flux Density

The physical realization of a transformer design aims to determine several critical parameters:
* **Core Size**: This includes the core cross-sectional area (`A_c`), the window area (`WA`), the mean length per turn (`MLT`), and the magnetic path length (`l_c`).
* **Number of Turns**: For each winding (`N_1`, `N_2`, ..., `N_k`).
* **Wire Size**: The cross-sectional area of the wire used for each winding.
* **Peak Flux Density (ΔB)**: This is a crucial parameter that must be carefully selected.
* `ΔB` influences both core losses and winding losses.
* It must be kept well below the saturation flux density (`B_sat`) of the core material to prevent the transformer from saturating.
* An optimal `ΔB` value is sought to minimize total losses.
* **Relationship between ΔB and Turns**: The peak flux density is related to the

number of turns and core area by the equation: `ΔB = λ_1 / (2 * N_1 * A_c)`, where `λ_1` is the flux linkage and `N_1` is the number of turns in the primary winding. This equation highlights the inverse relationship between `ΔB` and `N_1` for a given flux linkage and core area.

3. Window Area and Winding Allocation

Transformers typically have multiple windings (e.g., primary, secondary, auxiliary) that must be physically accommodated within the core's window area (`WA`).
* **Window Area Definition**: `WA` is the total cross-sectional area available for all windings. For an E-core, there are usually two windows, while a U-core or C-core might have one.
* **Fractional Allocation (`γ_j`)**: To manage the space, a fraction `γ_j` of the total window area is allotted to each winding `j`.
* The sum of all fractional allocations must equal one: `γ_1 + γ_2 + ... + γ_k = 1`.
* Each `γ_j` must be greater than 0 (as each winding needs some area) and less than 1 (unless it's the only winding).
* **Impact of `γ_j`**: The `γ_j` value directly influences the cross-sectional area of the wire that can be used for winding `j`, which in turn affects its resistance and copper loss.

4. Calculating Copper Loss and Winding Resistance

The copper loss in the `j`-th winding (`P_cu,j`) is given by `I_j^2 * R_j`, where `I_j` is the RMS current in winding `j` and `R_j` is its DC resistance.
* **Winding Resistance (`R_j`)**: Neglecting high-frequency AC effects (skin and proximity effects), the resistance is calculated as: `R_j = ρ * L_j / A_w,j`.
* `ρ` is the wire resistivity (material-dependent, e.g., copper, and temperature-dependent).
* `L_j` is the total length of wire used for winding `j`.
* `A_w,j` is the cross-sectional area of the wire used for winding `j`.
* **Length of Wire (`L_j`)**: The total length of wire for winding `j` is `L_j = N_j * MLT`, where `N_j` is the number of turns in winding `j` and `MLT` is the mean length per turn.
* **Cross-sectional Area of Wire (`A_w,j`)**: The effective cross-sectional area of the wire for winding `j` is determined by the allocated window area, the winding fill factor, and the number of turns: `A_w,j = (WA * K_u * γ_j) / N_j`.
* `WA` is the total window area.
* `K_u` is the winding fill factor.
* `γ_j` is the fraction of window area allotted to winding `j`.
* `N_j` is the number of turns in winding `j`.
* **Substituting into `R_j`**: By substituting `L_j` and `A_w,j` into the resistance

formula, we get: `R_j = ρ * N_j^2 * MLT / (WA * K_u * γ_j)`.
* **Total Copper Loss**: The total copper loss for all `k` windings is the sum of individual winding losses: `P_cu_total = Σ (P_cu,j) = Σ (I_j^2 * R_j)`.
* This can be expressed as: `P_cu_total = (ρ * MLT / (WA * K_u)) * Σ (N_j^2 * I_j^2 / γ_j)`.
* Using turns ratio (`N_j / N_1`), this becomes: `P_cu_total = (ρ * MLT * N_1^2 / (WA * K_u)) * Σ ((N_j/N_1)^2 * I_j^2 / γ_j)`.

5. Mean Length Per Turn (MLT) and Winding Fill Factor (Ku)

- **Mean Length Per Turn (MLT)**: This represents the average length of a single loop of wire around the core. For an E-core with a bobbin, it accounts for the straight sections and the curved sections where the wire wraps around the core's central leg and side legs.
- MLT is derived from the bobbin dimensions, considering the inner and outer layers of windings and the curvature. For an E-core, it can be approximated as `πA + 2D + 2E + 4I`, where `A` is the radius of curvature, `D` and `E` are bobbin dimensions, and `I` accounts for insulation thickness.
- **Winding Fill Factor (Ku)**: This factor accounts for the non-ideal packing of wires within the window area.
- `K_u` is always less than 1 because:
- Wires have enamel insulation, which takes up space.
- Circular conductors cannot perfectly fill a rectangular space, leaving gaps.
- Manufacturing tolerances and winding techniques also contribute to unused space.
- Typical values for `K_u`:
- 0.5 for low-voltage, round wire conductors.
- 0.65 for low-voltage foil windings.
- 0.1 for high-voltage transformer windings (due to thicker insulation requirements).

6. Optimization of Copper Losses (Gamma Optimization)

The total copper loss `P_cu_total` is a function of the allocated window area fractions `γ_j`.
* **Individual Winding Loss Behavior**:
* As `γ_j` approaches 0, the wire cross-section `A_w,j` becomes very small, leading to very high resistance and thus very high `P_cu,j`.
* As `γ_j` approaches 1, `A_w,j` becomes large, leading to low resistance and low

`P_cu,j`.
* **Total Copper Loss Curve**: When plotting `P_cu_total` against `γ_1` (assuming other `γ_j` values adjust accordingly), the curve typically shows a U-shape.
* If `γ_1` is very small, `P_cu,1` is high, and `P_cu_total` is high.
* If `γ_1` is very large (approaching 1), it means other `γ_j` values are small, making `P_cu,j` for `j ≠ 1` very high, again leading to high `P_cu_total`.
* There exists an **optimum `γ_j`** for each winding that minimizes the total copper loss. This requires an optimization process, potentially using linear optimization or advanced algorithms.

7. Optimization of Total Losses (Delta B Optimization)

The ultimate goal is to minimize the total power loss in the transformer, which is the sum of core loss and total copper loss: `P_total = P_core + P_cu_total`.
* **Relationship with ΔB**:
* `P_core` is proportional to `ΔB^β` (increases with `ΔB`).
* `P_cu_total` is inversely proportional to `ΔB^2` (decreases with `ΔB`), as `N_1` is inversely proportional to `ΔB`, and `P_cu_total` is proportional to `N_1^2`.
* **Total Loss Curve**: When `P_total` is plotted against `ΔB`:
* At low `ΔB`, `P_core` is low, but `P_cu_total` is high (due to many turns).
* At high `ΔB`, `P_core` is high, but `P_cu_total` is low (due to fewer turns).
* This results in a U-shaped curve for `P_total`, indicating an **optimum `ΔB`** where the total loss is minimal.
* **Finding Optimum ΔB**: The optimum `ΔB` can be found by differentiating `P_total` with respect to `ΔB` and setting the derivative to zero: `dP_total / dΔB = 0`. This implies that the rate of change of core loss with `ΔB` is equal to the negative rate of change of copper loss with `ΔB`.

8. Two-Level Optimization Strategy

Transformer design involves a two-level optimization process to achieve minimal total losses:
1. **First-level Optimization (Gamma Optimization)**: Determine the optimum `γ_j` values for each winding to minimize the total copper loss (`P_cu_total`). This ensures efficient utilization of the window area.
2. **Second-level Optimization (Delta B Optimization)**: Determine the optimum `ΔB` value that minimizes the overall `P_total` (sum of `P_core` and `P_cu_total`). This balances core and copper losses.

Once these optimum values are found, they can be used to calculate the required number of turns for the primary winding (`N_1`) and, subsequently, for other

windings using the turns ratios. This systematic approach allows for the physical realization of a high-frequency transformer with optimized performance.

Key Takeaways

- [ ] Transformer design involves balancing core losses (hysteresis, eddy currents) and winding losses (copper losses).
- [ ] Core loss is proportional to `ΔB^β`, while copper loss is inversely proportional to `ΔB^2`.
- [ ] The window area (`WA`) must accommodate all windings, with `γ_j` representing the fraction allocated to each winding `j`.
- [ ] The sum of all `γ_j` fractions must equal 1, and each `γ_j` must be between 0 and 1.
- [ ] Winding resistance depends on wire resistivity (`ρ`), total wire length (`L_j`), and wire cross-sectional area (`A_w,j`).
- [ ] `L_j` is calculated using the number of turns (`N_j`) and the mean length per turn (`MLT`).
- [ ] `A_w,j` is determined by `WA`, `γ_j`, `N_j`, and the winding fill factor (`K_u`).
- [ ] `K_u` accounts for non-ideal wire packing and insulation, typically being less than 1.
- [ ] Optimization is a two-step process: first, optimize `γ_j` for minimal copper loss, then optimize `ΔB` for minimal total loss.
- [ ] The optimum `ΔB` occurs where the total loss (core loss + copper loss) is at its minimum, resulting in a U-shaped loss curve.

Conclusion

The detailed exploration of transformer design parameters, from core material properties to winding configurations, provides a robust framework for engineers. By systematically analyzing core losses, winding losses, and the physical constraints imposed by the window area, we can move beyond theoretical understanding to practical realization. The two-level optimization strategy—first for window area allocation (`γ_j`) to minimize copper losses, and then for peak flux density (`ΔB`) to minimize total losses—is a powerful tool. This methodical approach ensures that the designed transformer not only meets the required electrical specifications but also operates with maximum efficiency and minimal thermal stress, which are critical considerations for the reliable and sustainable operation of modern charging infrastructure and other power electronic systems.